

Rules Of Thumb For Writing

Mittwoch, 25. Februar 2026 11:21

Reference: [Rules of thumb for Scientific Writing](#)

Overview:

No.	Section	Main Functions	Preferred Style	Remark
1	Title	• indicates content and main discoveries • attracts readers attention	• short and simple (7-10 words) • Purposive (aims at specific audience)	• Try to pick a catchy title • Avoid complex grammes • Avoid redundancy (An investigation of ...)
2	Abstract	• Reflects the main story of the RA • Calls attention but avoids extra explanations	• Past (perfect) tense and passive voice • Short and concise sentences • No citations, tables, equations, graphs, etc.	• Give the overall idea • What was done? • What was found? • What are the main conclusions? Bing summary 'numbers'
3	Introduction	• Introduces the topic and defines the terminology • Relates existing research • Indicates the focus of the paper and research objectives	• Simple tense for referring to established knowledge • Past tense for literature review	• Use state-of-the-art references • Define your terminology to avoid confusion • Move 1: • Introduce the topic and emphasize why it is important • Move 2: Relate to current knowledge: • What has been done? • What needs to be done? • Move 3: • Introduce your work • give the purpose -> What is the gap? • and main objective
4	Methodology	• Provide enough detail for competent researchers to repeat the experiment • Who, what, when, where, how, and why?	• Past tense but active voice • Correct and internationally recognised style and format (units, variables, materials, etc)	• Mention everything you did that can make importance to the results • Describe the experimental set-up • Explain used techniques • Object of the study • Establish an author's voice ("we decide to ignore this date") • RA is like a cook-book, be specific and provide all necessary detail • If a technique is familiar, only use its name (don't re-explain) • Use simplest example to explain complex methodology
6	Results	• Give summary results in graphics and numbers • Compare different treatments • gives quantified proofs (statistical tests)	• Past tense • use tables and graphs, and other illustrations	• Present summary data related to the RA objectives and not all research results • Give more emphasise on what should be emphasised • Call attention to the most significant findings • Make clear separation between yours and others work
7	Conclusions and Discussions	• Answers research questions / objectives • explains discrepancies and unexpected findings • states importance of discoveries and future implications	• Simple or present tense • Past tens if it is related to results • Allows scientific speculations if necessary	• Do not recapitulate results but make statements • Make strong statements (avoid "It may be concluded .." style) • Do not hide unexpected results - they can be the most important
8	References	• Gives list of related literature and information sources	• depends on journal but authors/editors, year and title must be included.	• Always cite the most accessible references • Cite primary source rather than review papers

Summary

Introduction

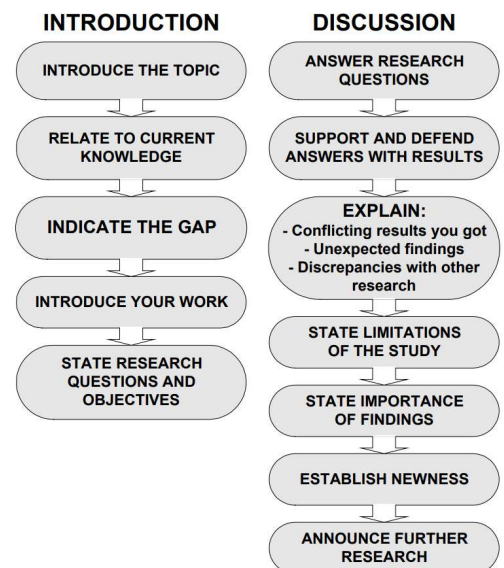
- Technical document that describes significant experimental, theoretical, or observational extension of current knowledge, or advanced in the practical application of known principles
- Articles should report on research findings that are
 - Sound -> valid
 - Previously unpublished -> original
 - Add some new understanding, observation, proofs, and potentially important information
- Has a required structure and style
 - Introduction
 - Methods
 - Results
 - Discussion

Methodology

- Describe experimental set-up
- Explain used techniques
- Object of the study
 - Standard Structure and Elements of an RA
 - macro. meso, and micro levels of a RA
 - general problems with readability and communication
 - Function and content of Introduction, Methods, Results and Discussion section
 - Writing successful abstracts
 - principles of submitting and publishing

Results

- Abbreviation research article sections (further referred to as RAS)
- Core body of an RA
 - Title, Authors, Abstract
 - Introduction
 - Methodology
 - Results
 - Conclusions and Discussion
 - References
- Research Article Elements:
 - Tables -> R, D
 - Figures (graphs, maps, diagrams, sketches) -> R, D
 - Equations -> I, M
 - Citations -> I, M
 - Footnotes
 - Comments
- Research Article (RA) sub structure
 - All sections can be separated in subsections or signposts
 - Subsections may be arbitrary but improve the structure of a RA
 - Recognized subsections
 - research topic and definitions
 - research objectives (questions)
 - methodological techniques
 - experimental set-up
 - objective of the study (study area)
 - Main discoveries (analysed data)
 - Answers on research questions, explanation of the conclusions, and further research implications.



Conclusion and Discussion

Name	Golden Rule
Take a readers view	Write for your audience and not for yourself
Tell a story	Direct your RA but keep a clear focus in the paper and present only results that relate to it
Be yourself	Write like you speak and then revise and polish
Make it simple	Use simplest example to explain complex methodology
Make it concrete	Use concrete words and strong verbs, avoid noun clusters (more than three words), abstract and ambiguous words

Make it short	Avoid redundancy, repetition and over-explanation of familiar techniques and terminology
Take responsibility	Make a clear distinction between your work and that of others
Make strong statements	"We concluded..." instead of "It may be concluded"
Be self-critical	Considering uncertainty of conclusions and their implications and acknowledge the work of others

Unexpected Findings

- Clear and coherent writing, rather than a fancy or complex, pseudo-scientific style.
- Information gain is especially enhanced by the use of examples
- Quality of an abstract is much more important
- Abstract should present the 'story' of the RA in miniature and should be readable standalone

WRITING A RA IN 40 STEPS

1. Make a working title
2. Introduce the topic and define terminology
3. Emphasize why the topic is important
4. relate to current knowledge: What's been done`?
5. Indicate the gap: what needs to be done?
6. Pose research question
7. Give purpose and objectives
8. List methodological steps
9. Explain theory behind the methodology used
10. Describe experimental set-up
11. Describe object of the study (technical details)
12. Give summary results
13. Compare different results
14. Focus on main discoveries
15. Answer research questions (conclusions=
16. Support and defend answers
17. Explain conflicting results, unexpected findings and discrepancies with other research
18. State limitations of the study
19. State importance findings
20. Establish newness
21. Announce further research